LAB 1

↳ 6 cells hidden

LAB 2

↳ 11 cells hidden

LAB 3

↳ 4 cells hidden

LAB 4

↳ 20 cells hidden

LAB 5

↳ 1 cell hidden

LAB 6

Email spam classification

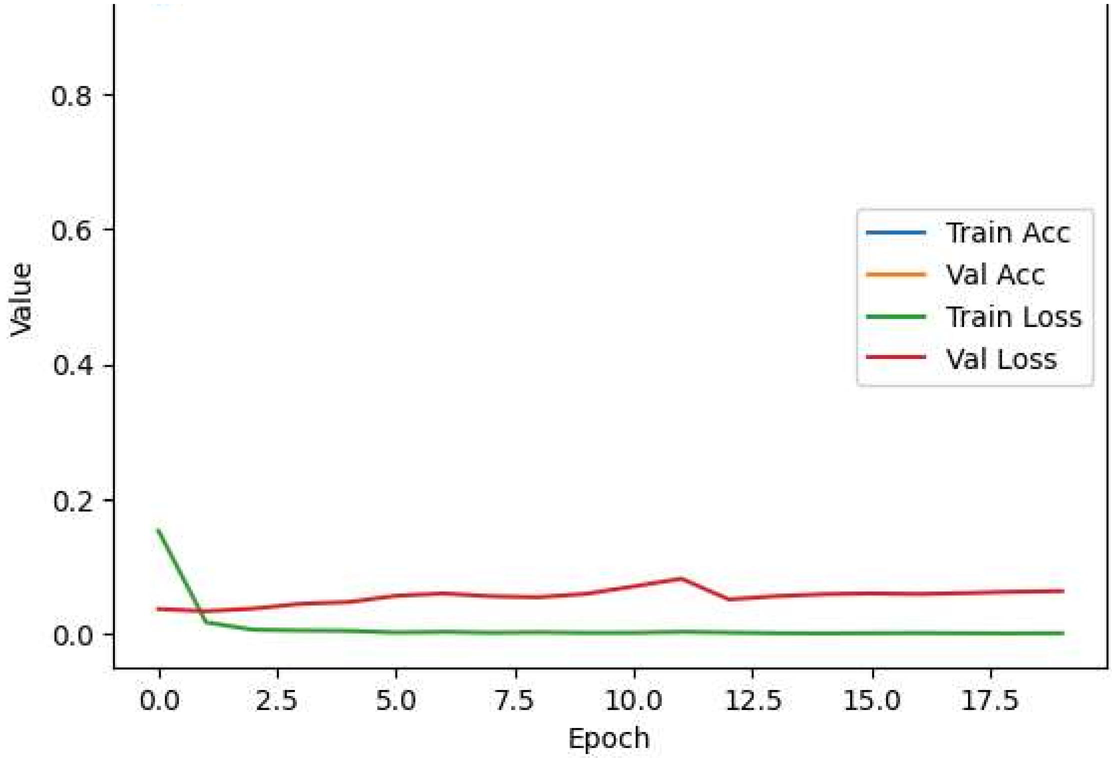
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_s
from sklearn.feature_extraction.text import TfidfVectorizer

from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam

# ================================
# 1️⃣ Load Dataset (CORRECT PATH)
# ================================

# Change this if needed after checking os.listdir('/content')
df = pd.read_csv('/content/spam_or_not_spam.csv')

print("Dataset Shape:", df.shape)
print(df.head())

# ================================
# 2️⃣ Preprocessing
# ================================

X_text = df['email'].fillna('')
y = df['label'].astype('float32')

vectorizer = TfidfVectorizer(max_features=1000)
X = vectorizer.fit_transform(X_text).toarray().astype('float32')

# Train/Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# ================================
# 3️⃣ Build Model
# ================================

def create_model(lr):
    model = Sequential([
        Dense(64, activation='relu', input_shape=(X.shape[1],)),
        Dense(32, activation='relu'),
        Dense(1, activation='sigmoid')
    ])

    model.compile(
        optimizer=Adam(learning_rate=lr),
        loss='binary_crossentropy',
        metrics=['accuracy']
    )

    return model

# ================================
# 4️⃣ Training + Graphs
```

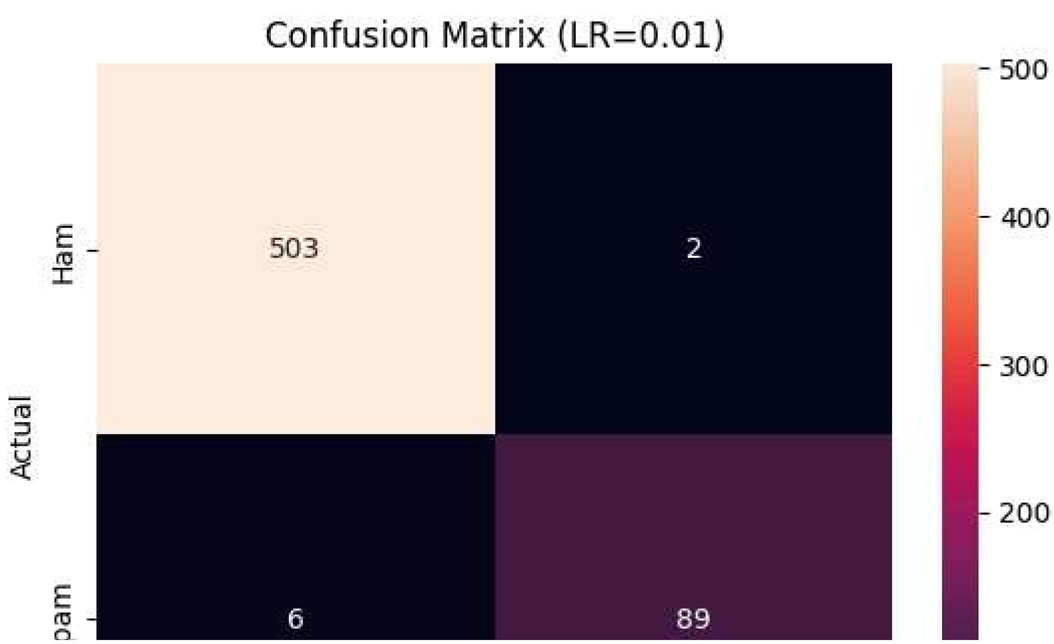
```
# ===============================

learning_rates = [0.01, 0.001, 0.005]

for lr in learning_rates:

    print("\n===============================")
    print("Training with LR =", lr)

    model = create_model(lr)

    history = model.fit(
        X_train, y_train,
        epochs=20,   # reduced for stability
        batch_size=32,
        validation_split=0.1,
        verbose=1
    )

    # 📊 Accuracy & Loss Graph
    plt.figure()
    plt.plot(history.history['accuracy'])
    plt.plot(history.history['val_accuracy'])
    plt.plot(history.history['loss'])
    plt.plot(history.history['val_loss'])
    plt.title(f"Training vs Validation (LR={lr})")
    plt.xlabel("Epoch")
    plt.ylabel("Value")
    plt.legend(["Train Acc", "Val Acc", "Train Loss", "Val Loss"])
    plt.show()

    # ===============================
    # Evaluation
    # ===============================

    y_pred = (model.predict(X_test) > 0.5).astype("int32")

    print("Accuracy :", accuracy_score(y_test, y_pred))
    print("Precision:", precision_score(y_test, y_pred))
    print("Recall   :", recall_score(y_test, y_pred))
    print("F1 Score :", f1_score(y_test, y_pred))
    print("\nClassification Report:\n", classification_report(y_test, y_pred))

    # ===============================
    # Confusion Matrix
    # ===============================

    cm = confusion_matrix(y_test, y_pred)

    plt.figure()
    sns.heatmap(cm, annot=True, fmt='d',
```

```
                xticklabels=['Ham', 'Spam'],
                yticklabels=['Ham', 'Spam'])
        plt.title(f"Confusion Matrix (LR={lr})")
        plt.xlabel("Predicted")
        plt.ylabel("Actual")
        plt.show()
```

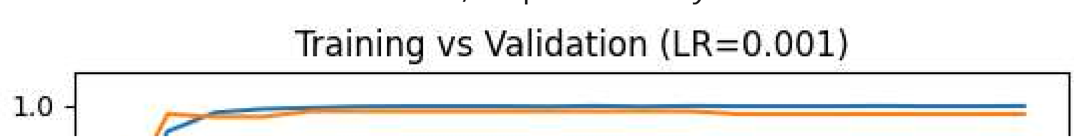
```
Dataset Shape: (3000, 2)
                                                email  label
0   date wed NUMBER aug NUMBER NUMBER NUMBER NUMB...      0
1  martin a posted tassos papadopoulos the greek ...      0
2  man threatens explosion in moscow thursday aug...      0
3  klez the virus that won t die already the most...      0
4   in adding cream to spaghetti carbonara which ...      0

==============================
Training with LR = 0.01
Epoch 1/20
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserW
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
68/68 ━━━━━━━━━━━━━━━━━━━━ 4s 21ms/step - accuracy: 0.8910 - loss: 0.2836 - val_
Epoch 2/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 10ms/step - accuracy: 0.9950 - loss: 0.0144 - val_
Epoch 3/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9978 - loss: 0.0070 - val_a
Epoch 4/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.9994 - loss: 0.0019 - val_a
Epoch 5/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9993 - loss: 0.0034 - val_a
Epoch 6/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9999 - loss: 0.0015 - val_a
Epoch 7/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9993 - loss: 0.0017 - val_a
Epoch 8/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9994 - loss: 0.0022 - val_a
Epoch 9/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9995 - loss: 0.0016 - val_a
Epoch 10/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9997 - loss: 9.9309e-04 - v
Epoch 11/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9998 - loss: 5.4585e-04 - v
Epoch 12/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9995 - loss: 0.0020 - val_a
Epoch 13/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9992 - loss: 0.0033 - val_a
Epoch 14/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9999 - loss: 4.2332e-04 - v
Epoch 15/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9998 - loss: 3.3104e-04 - v
Epoch 16/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9996 - loss: 5.4753e-04 - v
Epoch 17/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.9994 - loss: 0.0012 - val_a
Epoch 18/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9986 - loss: 0.0016 - val_a
Epoch 19/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9997 - loss: 5.4544e-04 - v
Epoch 20/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9991 - loss: 9.7037e-04 - v
```

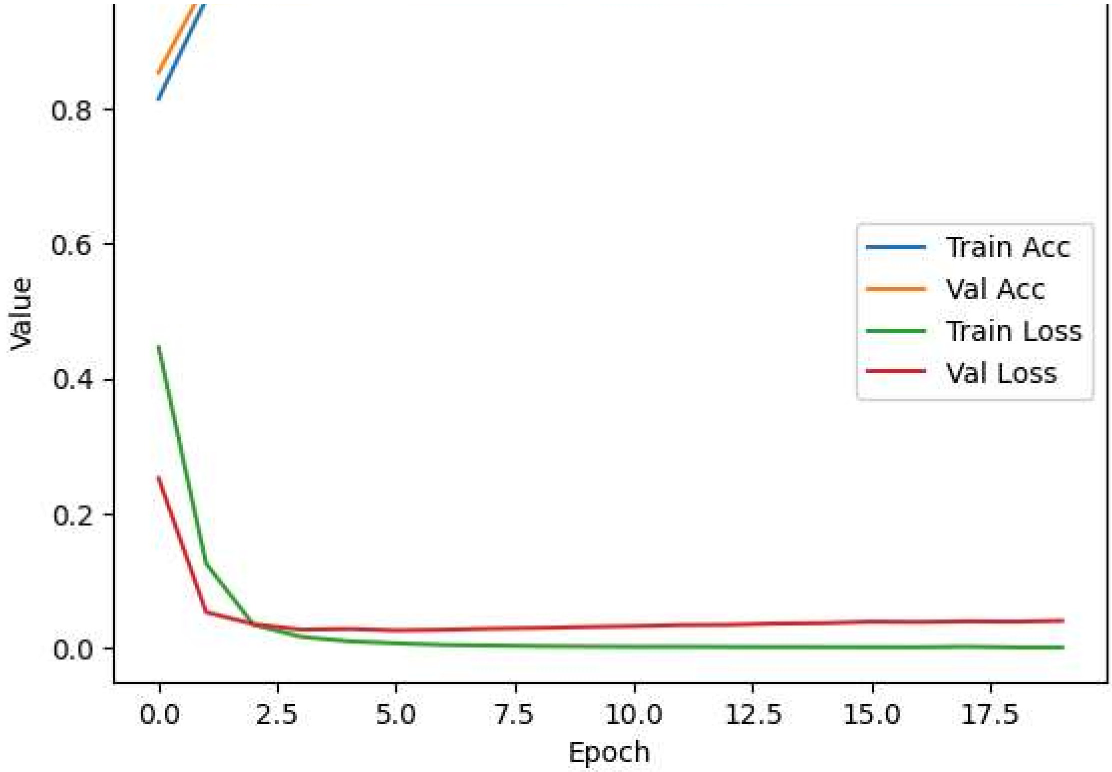



```
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Accuracy : 0.9866666666666667
Precision: 0.978021978021978
Recall   : 0.9368421052631579
F1 Score : 0.956989247311828

Classification Report:
              precision    recall  f1-score   support

         0.0       0.99      1.00      0.99       505
         1.0       0.98      0.94      0.96        95

    accuracy                           0.99       600
   macro avg       0.98      0.97      0.97       600
weighted avg       0.99      0.99      0.99       600
```

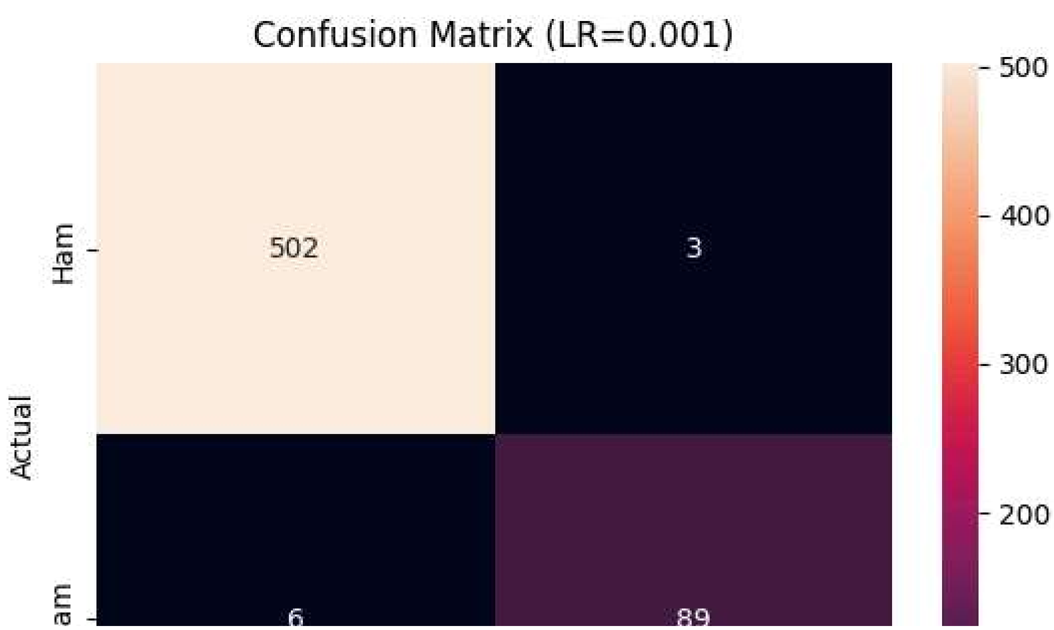



```
==============================
Training with LR = 0.001
Epoch 1/20
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserW
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
68/68 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - accuracy: 0.7516 - loss: 0.5667 - val_a
Epoch 2/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9305 - loss: 0.1706 - val_a
Epoch 3/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9904 - loss: 0.0359 - val_a
Epoch 4/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9896 - loss: 0.0246 - val_a
Epoch 5/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 0.9969 - loss: 0.0088 - val_a
Epoch 6/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9986 - loss: 0.0078 - val_a
Epoch 7/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9981 - loss: 0.0062 - val_a
Epoch 8/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.9993 - loss: 0.0034 - val_a
Epoch 9/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9995 - loss: 0.0025 - val_a
Epoch 10/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9983 - loss: 0.0025 - val_a
Epoch 11/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9998 - loss: 0.0018 - val_a
Epoch 12/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 0.9978 - loss: 0.0031 - val_a
Epoch 13/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 1s 8ms/step - accuracy: 1.0000 - loss: 0.0018 - val_a
Epoch 14/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9993 - loss: 0.0013 - val_a
Epoch 15/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9989 - loss: 0.0021 - val_a
Epoch 16/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9993 - loss: 0.0011 - val_a
Epoch 17/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 7.1674e-04 - v
Epoch 18/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9978 - loss: 0.0037 - val_a
Epoch 19/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9997 - loss: 7.4142e-04 - v
Epoch 20/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9998 - loss: 0.0011 - val_a
```






```
19/19 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step
Accuracy : 0.985
Precision: 0.967391304347826
Recall   : 0.9368421052631579
F1 Score : 0.9518716577540107

Classification Report:
              precision    recall  f1-score   support

         0.0       0.99      0.99      0.99       505
         1.0       0.97      0.94      0.95        95

    accuracy                           0.98       600
   macro avg       0.98      0.97      0.97       600
weighted avg       0.98      0.98      0.98       600
```




```
==============================
Training with LR = 0.005
Epoch 1/20
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserW
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
68/68 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - accuracy: 0.8431 - loss: 0.3794 - val_a
Epoch 2/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9923 - loss: 0.0236 - val_a
Epoch 3/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9991 - loss: 0.0046 - val_a
Epoch 4/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9984 - loss: 0.0034 - val_a
Epoch 5/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9975 - loss: 0.0056 - val_a
Epoch 6/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9997 - loss: 0.0013 - val_a
Epoch 7/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9981 - loss: 0.0044 - val_a
Epoch 8/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9976 - loss: 0.0035 - val_a
Epoch 9/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9996 - loss: 0.0019 - val_a
Epoch 10/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9998 - loss: 5.6092e-04 - v
Epoch 11/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9997 - loss: 0.0021 - val_a
Epoch 12/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9993 - loss: 0.0011 - val_a
Epoch 13/20
68/68 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9999 - loss: 5.9921e-04 - v
Epoch 14/20
```